Isometry Pursuit: Convex Recovery of Orthonormal Column Submatrices

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Abstract

Isometry pursuit is a convex algorithm for identifying orthonormal column-submatrices of wide matrices. It consists of a vector normalization followed by multitask basis pursuit. Applied to Jacobians of putative coordinate functions, it helps identity locally isometric embeddings from within interpretable dictionaries. We provide theoretical and experimental results justifying this method, including a proof with realistic assumptions that such isometric submatrices, should they exist, are contained within the obtained support. For problems involving coordinate selection and diversification, it offers a synergistic alternative to greedy and brute force search.

1 Introduction

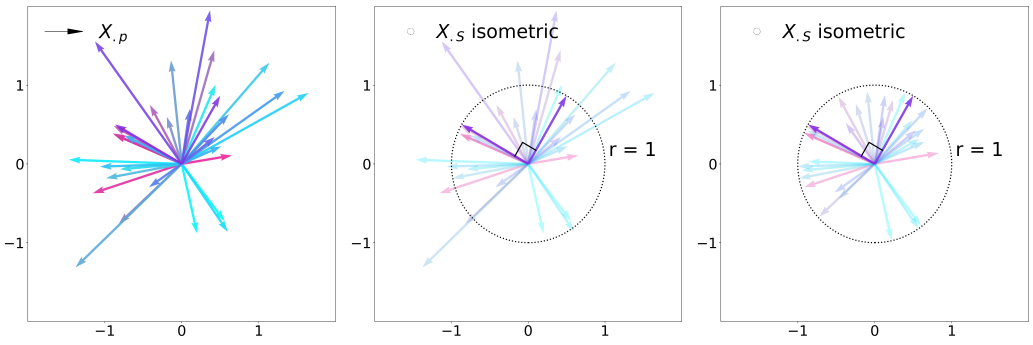
Many real-world problems may be abstracted as selecting a subset of the columns of a matrix representing stochastic observations or analytically exact data. This paper focuses on a simple such problem. Given a rank D matrix $X \in \mathbb{R}^{D\times P}$ with $P > D$, select a square submatrix $X_{\cdot S}$ where subset $S \subset [P]$ satisfies $|S| = D$ that is as isometric as possible.

This problem arises in feature learning because while the coordinate functions of a given feature space may have no intrinsic meaning, it is often possible to generate a dictionary of interpretable features for consideration as potential parametrizing coordinates. When this is the case, selection of candidate interpretable features as coordinates can take the above form. Isometric embeddings - data mappings that compress high-dimensional distributions into low dimensions while preserving distances and angles on the data surface - have differentials that are isometric matrices. Thus, selecting isometric embeddings from without dictionaries of putative coordinates can be cast as a linear isometric submatrix coordinate selection problem. Identification of such coordinates generally aids mechanistic understanding, generative control, and statistical efficiency.

This paper takes an optimization perspective on the problem of selecting isometric submatrices. The background develops the the problem from the perspective of non-linear embeddings. However, the focus switches to a purely linear algebraic perspective in the remainder of the paper.

The main contribution of this paper is to show that an adapted version of the multitask basis pursuit feature learning algorithm in Koelle et al. [1] is a convex alternative isometry selection algorithm. This enables identification of low-dimensional coordinates within a Jacobian that preserve the original geometry of the data manifold under mapping given by the flow lines This algorithm - IsometryPursuit - is multitask basis pursuit applied to an appropriately normalized X selects orthonormal submatrices. Given vectors in $\mathbb{R}^D$, the normalization log-symmetrizes length and favors those closer to unit length, while basis pursuit favors those which are orthogonal. Our theoretical results formalize this intuition, and show the usefulness of IsometryPursuit as a trimming procedure prior to brute force search for diversification and interpretable coordinate selection. It is an alternative to greedy coordinate selection approaches such as those in Cai and Wang [2], Chen and Meila [3], Kohli et al. [4], Jones et al. [5].

[1]Work conducted outside of Amazon.



(a) Raw data (b) Isometric subset (c) Normalization step

Figure 1: 1a Raw data X 1b An isometric subset 1c Our method - vectors of length 1 maintain longest length after normalization

2 Background

Our goal is, given a matrix $X \in \mathbb{R}^{D \times P}$, to select a subset $S \subset [P]$ with $|S| = D$ such that $X_{.S}$ is as orthonormal as possible in a computationally efficient way. Figure 1a and 1b provide a visual depiction of this problem when $D = 2$ and $P = 30$. We motivate this problem from the perspective of non-linear embeddings.

2.1 Isometry

Isometries are a classical object in numerical linear algebra and geometry. They preserve distances and angles, and such can be used faithfully for subsequent human or algorithmic analysis. This motivates embedding algorithms such as Local Tangent Space Alignment [6], Multidimensional Scaling [7], and Isomap [8].

Definition 1 *A matrix $X \in \mathbb{R}^{D \times D}$ is an **isometry** or, equivalently, **unitary** if $XX^T = I_D$.*

Proposition 1 *The singular values $\sigma_1(X) \dots \sigma_D(X)$ are equal to 1 if and only if X is an isometry.*

Proposition 2 *If $P = D$, the component vectors $X_{.1} \dots X_{.D} \in \mathbb{R}^D$ form an isometry if and only if, for all $d, d' \in [D], \langle X_{.d}, X_{.d'} \rangle = \begin{cases} 1 & d = d' \\ 0 & d \neq d' \end{cases}$.*

Definition 2 *The **differential** of a smooth map $\phi : \mathcal{M} \to \mathcal{N}$ between D dimensional manifolds $\mathcal{M} \subseteq \mathbb{R}^B$ and $\mathcal{N} \subseteq \mathbb{R}^P$ is a map in tangent bases $x_1 \dots x_D$ of $T_\xi \mathcal{M}$ and $y_1 \dots y_D$ of $T_{\phi(\xi)} \mathcal{N}$ consisting of entries*

$$D\phi(\xi)_{ij} = \langle y_i, \frac{\partial \phi}{\partial x_j} \rangle (\xi) \tag{1}$$

Definition 3 *A map ϕ between D dimensional submanifolds is an **isometry at a point** $\xi \in \mathcal{M}$ if $D\phi(\xi)$ is an isometry.*

Definition 4 *A map ϕ between D dimensional submanifolds is **conformal at a point** $\xi \in \mathcal{M}$ if there exists a scalar $\lambda(\xi) > 0$ such that $D\phi(\xi)^T D\phi(\xi) = \lambda(\xi) I_D$.*

Proposition 3 *Let $\phi : \mathcal{M} \to \mathcal{N}$ be a map between D dimensional submanifolds.*

$$d_{\mathcal{N}}(\phi(\xi), \phi(\xi')) = d_{\mathcal{M}}(\xi, \xi'), \tag{2}$$

if and only if ϕ is smooth and is an isometry at every point [9, Problem 6-7].

Reference	Algorithm	Variable	Loss	Linearized	Domain	Identifies
Chen and Meila [3]	GREEDY	$S \in \binom{[P]}{M}$	$-\frac{1}{N}\sum_{n=1}^{N} \log \frac{\det\left(x_S^T x_S\right)^{1/2}}{\prod_{s\in S}\|x_S\|_2}(\xi_n)$	Yes	Dataset	Conformal
Kohli et al. [4]	GREEDY	$S \in \binom{[P]}{D}$	$\sup_{\xi',\xi''\in\mathcal{N}(\xi)\ \xi'\neq\xi''} r_S(\xi',\xi'') \sup_{\xi',\xi''\in\mathcal{N}(\xi)\ \xi'\neq\xi''} r_S(\xi',\xi'')^{-1}$ where $r_S(\xi',\xi'') := \frac{\|f_{.S}(\xi') - f_{.S}(\xi'')\|_2}{\|\xi' - \xi''\|_2}$.	No	Point	Conformal
Koelle et al. [1]	FISTA	$\beta \in \mathbb{R}^{N\times P\times D}$	$\frac{1}{2}\sum_{n=1}^{N}\|I_d - Df(\xi_n)\beta_{n..}\|_F^2 + \lambda\sqrt{ND}\sum_{p=1}^{P}\left(\sum_{n=1}^{N}\sum_{d=1}^{D}\beta_{npd}^2\right)^{1/2}$	Yes	Dataset	Bijection

Table 1: Dictionary-based manifold learning losses. Note that we set the eigenfunction penalty to 0 for Chen and Meila [3] to emphasize the geometric condition

2.2 Coordinate selection

More recently, methods for selecting D coordinates from a dictionary $f_{1:P}$ of candidate features have emerged [4, 10**?**]. Loss functions for isometry vary along many axes. For example, functions may be measured at a single point [4] or over an entire dataset [10, 1, 3]. The loss function may make use of either linearized or non-linearized objectives. Objectives can be computed analytically or from numerical methods. Isometry may be relaxed as optimization target to the weaker notion of conformalness or embedding.

The dictionary-based coordinate search problem best subset selection problem is

$$\arg\min_{S\in\binom{[P]}{D}} \ell(f_{.S}) \tag{3}$$

where ℓ is some loss function that measures isometricness of a selected functional subset $f_{.S}$. Often, this problem is linearized via defining a loss at a point ξ as

$$\ell = \ell'(Df_{.S}(\xi)) \tag{4}$$

where ℓ' is now a loss function that measures the orthonormalness of the matrix $Df_{.S}(\xi)$. Non-linear objectives are often gradient estimators of linearized objectives, and so the distinction is smaller than it may appear.

Proposition 4 *The distortion loss function of f_S is equal to the condition number [11, 12] $\frac{\sigma_1(X)}{\sigma_D(X)}$ of $Df_{.S}(\xi)$ as $\mathcal{N}(\xi) \to \{\xi\}$.*

Proposition 5 *This is minimized by conformal maps.*

Furthermore, linearization can have computational advantages. For example, Koelle et al. [1] computes $Df_{.S}(\xi)$ via automatic differentiation.

2.3 Isometry losses

So far in this section, we have established the notion and desireableness of isometric embedding, and reviewed algorithms used for selecting column-submatrices that for conditions weaker than isometry. We now turn our review to the linear algebra literature.

Property 1 (Rotation invariance) *A matrix loss ℓ is rotation invariant if, for every orthonormal $U \in \mathbb{R}^{D\times D}$,*

$$\ell(UX) = \ell(X). \tag{5}$$

Property 2 (Unique isometry minimum) *A matrix loss ℓ is uniquely minimized by isometries if*

$$\arg\min_X \ell(X) = \{X : XX^T = I_D\}. \tag{6}$$

Loss Function	Definition	Unique min.	At isometry	Unit length	Ortho-gonal
Distortion [4]	$\sigma_{\max}(X)/\sigma_{\min}(X) - 1$	No	Yes	No	Yes
Generic spectral loss	$\ell(X) = \sum_{d=1}^{D} g(\sigma_d(X))$	Depends on g	Depends on g	Depends on g	Depends on g
Normalized determinant volume	$\left\lvert\det\left(n_q^{\times D}(X)\right)\right\rvert^{-1} - 1$	No	Yes	Yes	Yes
Frob. pseudoinverse	$\lVert n_q^{\times P}(X)^\dagger\rVert_F - \sqrt{D}$	No	Yes	Yes	Yes
IsometryPursuit	$\lVert n_q^{\times P}(X)^\dagger\rVert_{1,2} - D$	Yes	Yes	Yes	Yes

Table 2: Matrix loss functions evaluated for isometry pursuit. We evaluate four key properties: whether the loss has a unique minimum, whether it is minimized specifically at orthonormal matrices (isometries), whether it prefers unit-length vectors over scaled versions, and whether it prefers orthogonal (orthonormal) structure.

Definition 5 *Sra [13]* The **S-divergence** $D(P, P')$ on positive definite matrices P, P' is given by

$$D(P, P') = (\log\det(\frac{P+P'}{2}) - \frac{1}{2}\log\det(PP'))^{\frac{1}{2}}. \tag{7}$$

Proposition 6 *If X^TX is positive definite, then*

$$D(X^TX, I_D)^2 = \sum_{d=1}^{D} \log\frac{\sigma_d(X) + \sigma_d(X)^{-1}}{2}. \tag{8}$$

Proposition 7 *If X^TX is positive definite, then $D(X^TX, I_D)^2$ is uniquely minimized by isometric matrices.*

See [14] for a more extensive discussion of other matrix divergences. In Section 3, we will examine matrix loss functions capable of selecting isometric submatrices based on their singular values from a generic perspective.

2.4 Sparse convex coordinate selection

Equipped with a function ℓ that is minimized by an isometry, objective functions like 3 may be optimized in a variety of ways. Brute force algorithms consider all possible solutions are guaranteed to find the global minimum, but have the often prohibitive time complexity $O(C_\ell P^D)$ where C_ℓ is the cost of evaluating ℓ. In contrast, greedy algorithms consist of iteratively adding one element at a time to S. This algorithms have time complexity $O(C_\ell PD)$ and so are computationally more efficient than brute force algorithms, but can get stuck in local minima. Formal definitions are given in Section 6.1.

Sometimes, it is possible to introduce an objective which convexifies problems of the above form [15, 16, 17].

$$\arg\min_{\beta\in\mathbb{R}^P : y = X\beta} \rho(\beta) \tag{9}$$

for an overcomplete regression problem $y = X\beta$ is a classic example [18]. Overcomplete regression problems are distinguished by the fact that $P > D$ and so the solution β is not unique. When $\rho(\beta) = \lVert\beta\rVert_0$, this non-convex problem searches directly for a sparsest representing subset. However, when $\rho(\beta) = \lVert\beta\rVert_1$, the problem is a convex relaxation and may be solved efficiently e.g. via interior-point methods. Because the ℓ_1 objective is scale-sensitive, this relaxation is geometrically meaningful for coordinate selection only after normalizing the columns of X so that sparsity is not confounded with column length. See Schneider and Tardivel [19] for a recent discussion.

3 Method

The contributions of this paper is that we introduce an ℓ uniquely minimized by isometries that is convex in its parameters, induces sparsity, and is more appropriate than S-divergence

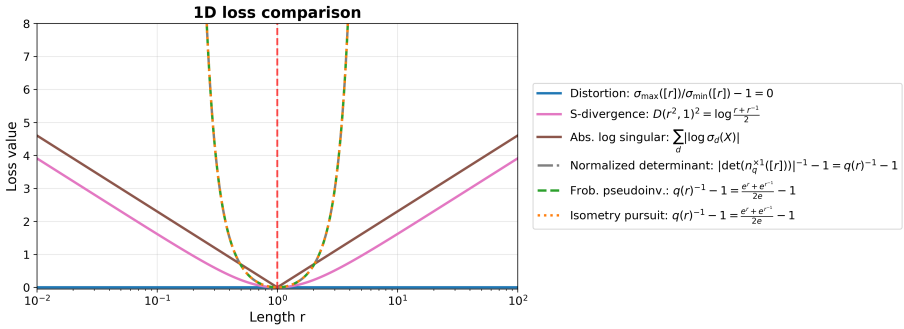


Figure 2: One-dimensional candidate loss curves for $X = [[r]]$, showing loss behavior versus vector length. The red vertical dashed line marks $r = 1$.

when an isometric submatrix is not present in that it always favors vectors of unit length. We call this ℓ multitask basis pursuit rather than group basis pursuite [10, 20] since grouping is dependent only on the structure of matrix-valued response variable I_D. We show that this approach, used to select independent dictionary elements in Koelle et al. [10, 1], is in fact capable of usefully solving the more specific problem of selecting pointwise isometries from a dictionary. We justify this approach in two ways. First, we should that this loss and its *method of frames* analog have intuitively preferable geometric properties to S-divergence and other singular-value based losses. Second, we exhibit the usefulness of this basis pursuit approach for convexifying the coordinate selection problem.

3.1 The problem with singular values

$D(X^T X, I_D)$ is not the only function of X minimized uniquely by orthonormal matrices and invariant under rotation. Consider

$$\ell : \mathbb{R}^{D \times P} \to \mathbb{R}^+ \tag{10}$$

$$X \mapsto \sum_{d=1}^{D} g(\sigma_d(X)) \tag{11}$$

where

$$g : \mathbb{R}^+ \to \mathbb{R}^+ \tag{12}$$

$$t \mapsto \log \frac{t + t^{-1}}{2}. \tag{13}$$

Clearly, any g uniquely minimized at 1 will provide a suitable rescaling of singular values. However, there are some nice properties we may also want.

Property 3 (Monotonicity away from one)

$$g'(t) \begin{cases} < 0 \ : \ 0 < t < 1 \\ = 0 \ : \ t = 1 \\ > 0 \ : \ t > 1. \end{cases} \tag{14}$$

Property 4 (Log-symmetry)

$$g(t) = g(1/t). \tag{15}$$

Property 5 (Column-rescaling monotonicity)

$$\frac{\partial \ell}{\partial \|x_{.d}\|_2} \begin{cases} > 0 \ : \ \|x_{.d}\|_2 > 1 \\ = 0 \ : \ \|x_{.d}\|_2 = 1 \\ < 0 \ : \ \|x_{.d}\|_2 < 1 \end{cases} \tag{16}$$

for all X.

Unfortunately, we show the following proposition.

Proposition 8 *There does not exist a g such that $\ell = \sum_{d=1}^{D} g(\sigma_d(X))$ such that ℓ satisfies Property 5 and Property 4.*

A suitable matrix for which no g can be chosen such that a log det divergence satisfies Properties **??** and **??** is shown in Section 6.2.2. Figure 2 gives graphs of candidate losses when $D = 1$. A view of the loss landscape for this matrix is shown in Figure 4. The level sets of the log det losses are not tangent to length at $r = 1$. This means that given an angle loss can actually be decreased by shrinking a vector away from length 1. This is an intuitively bad property for a loss that is supposed to prefer orthonormal matrices.

3.2 Normalization

Although starting from singular values is bad in the sense of Proposition , other losses do satisfy this property. These losses arise from the intuition that we can shrink vectors away from length 1 in a manner similar to how we selected g. Then, we can apply some loss function to the rescaled matrix that satisfies Property 5.

As mentioned in Section 2.4, it is important to normalize the columns of x before applying basis pursuit. A similar insight can be applied to the problem of selecting isometric submatrices. Since basis pursuit methods tend to select longer vectors, selection of orthonormal submatrices requires normalization such that both long and short candidate basis vectors are penalized in the subsequent regression. We introduce the following definition.

Definition 6 *Let $g : \mathbb{R}^+ \to \mathbb{R}^+$ satisfy Properties 3 and 4. We define a* ***symmetric normalization***

$$n_g : \mathbb{R}^D \setminus \{0\} \to \mathbb{R}^D \tag{17}$$

$$v \mapsto \frac{g(\|v\|_2)}{\|v\|_2} v. \tag{18}$$

We suggest to choose g as follows.

$$g : \mathbb{R}^+ \to \mathbb{R}^+ \tag{19}$$

$$t \mapsto \frac{2e}{e^t + e^{t^{-1}}}. \tag{20}$$

Besides satisfying the conditions in Definition 6, this choice of g has some additional nice properties.

Proposition 9 *For g defined in* (20)*, $-\log g$ is convex.*

Proposition 10 *The function g defined in* (20) *is smooth on $\mathbb{R}^+$.*

Proposition 11 *For g defined in* (20)*, $\log g(t)$ decays linearly in t as $t \to \infty$ and in t^{-1} as $t \to 0$.*

Proposition 12 *The inverse normalization $1/g$ in* (20) *is the normalized exponential of the LogSumExp [12] relaxation of* $\max(t, t^{-1})$*.*

We plot the effect of this normalization on our example data in Figure 1c.

3.3 Normalized trace loss

Many matrix functions applied to suitably normalized matrices are uniquely minimized by orthonormal matrices. Two examples are the determinant and also the Frobenius norm of the inverse.

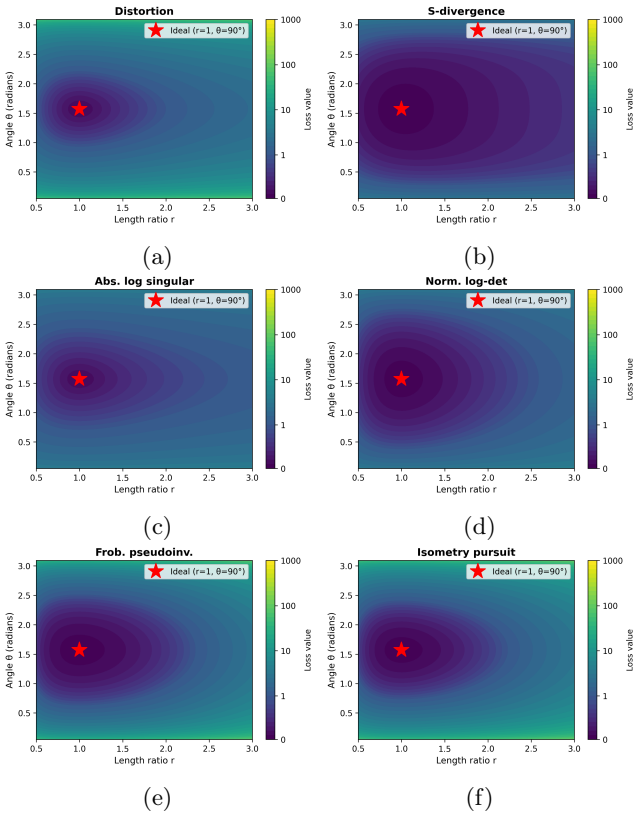


Figure 3: Loss function comparison for 2×2 matrices. Each landscape varies the length ratio r and angle θ between two column vectors. S-divergence and absolute log singular value are shown as representative generic spectral losses. The red star marks the ideal orthonormal setting $(r, \theta) = (1, \pi/2)$. See Table 2 for mathematical definitions.

Proposition 13 *The determinant of the normalized matrix* $\propto \sum_{d=1}^{D} \log \sigma_d$.

Proposition 14 *The Frobenius norm of the inverse* $= \sum_{d=1}^{D} \frac{1}{\sigma_d^2} \propto \sum_{d=1}^{D} \sigma_d$

Proposition 15 *The determinant of the normalized matrix satisfies Properties 3, 4, and 5.*

Proposition 16 *The Frobenius norm of the inverse Properties 3, 4, and 5.*

Unfortunately, we haven't identified a justification for one of these losses over the other beyond taste. The determinant will more gently penalize highly divergent singular values, but be less forgiving of small deviations from orthogonality. Figure 3 gives the analogous two-parameter view for $D = 2$ matrices, and Figure 4 extends this comparison to an equicorrelated $D = 3$ slice.

We have established that there are losses ℓ that satisfy our conditions for a good loss function on square matrices $X \in \mathbb{R}^{D \times D}$. However, the computational identification of such matrices is combinatorial in nature. Thus, in the actual application setting of selecting combinations of columns from a larger matrix $X \in \mathbb{R}^{D \times P}$, we still need to perform many expensive evaluations of singular values or inversions.

To fix this, we follow the well-trodden path of convexifying combinatorial problems by relaxing the problem to a convex optimization problem. As mentioned in the previous section, the Frobenius norm of the inverse satisfies our desiderata. This loss has a natural relaxation to the high-dimensional setting, namely the Frobenius norm of the Moore-Penrose pseudoinverse, also known as the method-of-frames solution. From here, there is an obvious extensive to the sparse basis pursuit objective.

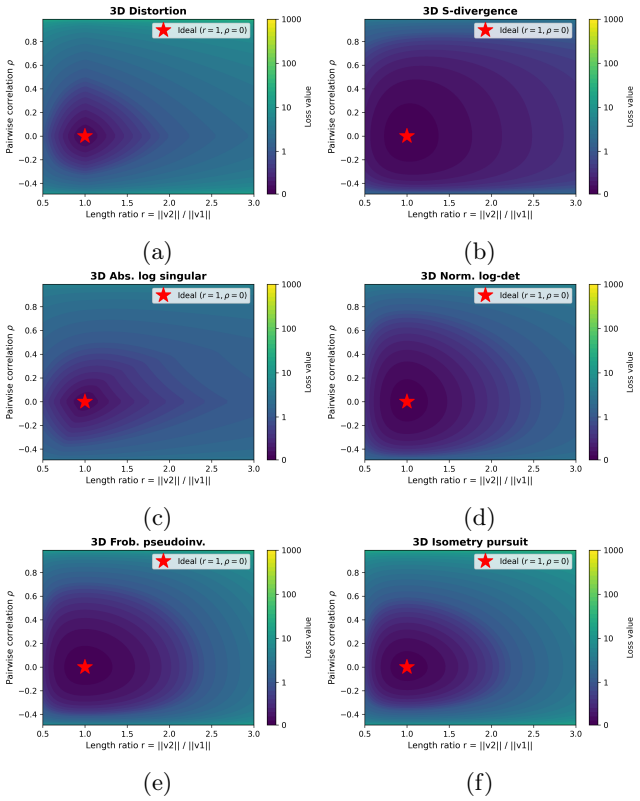


Figure 4: Loss function comparison for 3D equicorrelated matrices, shown as two-dimensional landscapes. Each matrix consists of three vectors in $\mathbb{R}^3$ with lengths $(1, r, 1)$ and common pairwise correlation ρ. The horizontal axis varies the length ratio $r = \|v_2\|_2/\|v_1\|_2$, while the vertical axis varies ρ. S-divergence and absolute log singular value are shown as representative generic spectral losses. The red star marks the ideal orthonormal setting $(r, \rho) = (1, 0)$.

3.4 Isometry pursuit

Isometry pursuit is the application of multitask basis pursuit to the normalized design matrix $w(X)$ to identify submatrices of X that are as orthonormal as possible. Define the multitask basis pursuit penalty

$$\|\cdot\|_{1,2} : \mathbb{R}^{P\times D} \to \mathbb{R}^+ \tag{21}$$

$$\beta \mapsto \sum_{p=1}^{P} \|\beta_{p\cdot}\|_2. \tag{22}$$

Given a matrix $Y \in \mathbb{R}^{D\times D}$, the multitask basis pursuit solutions are

$$\widehat{\beta}_{MBP}(X, Y) := \arg \min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ Y = X\beta. \tag{23}$$

The multitask basis pursuit solution is often not unique. See Schneider and Tardivel [21] for more details on uniqueness of basis pursuit solutions, for which the uniqueness condition is less strict than in the multitask case. We therefore define

$$\widehat{\beta}^{\ell_2}_{MLP}(X, Y) := \arg\min \|\beta\|_F : \beta \in \beta_{MBP}(X, Y). \tag{24}$$

By the strong convexity of the ℓ_2 norm, $\widehat{\beta}^{\ell_2}_{MLP}(X, Y)$ is well defined.

This implicit regularization assumption is reasonable. For example, in Tibshirani [22] the implicit selection of the minimum ℓ_2 norm solution among the lasso solutions was proven for the Least Angle Regression Selection (LARS) algorithm for solving the lasso, while

Mishkin and Pilanci [23] assumes that the group lasso solution is in fact min-norm prior to subsequent theoretical analyses. Empirically, the Splitting Conic Solver type method (which is itself a variant of the Alternating Direction Method of Multilpliers) [24] stably estimates the min-norm solution when initialized at 0, but besides a brief exploration in Section **??** we leave theoretical and experimental proof of this feature of the feature our optimization approach for future work, and instead make the ability of our optimizer to obtain the min-norm solution an assumption of our eventual proposition.

Isometry pursuit is then given by

$$\widehat{\beta}(X) := \widehat{\beta}^{\ell_2}_{MBP}(w(X), I_D) \tag{25}$$

where I_D is the D dimensional identity matrix and recovered functions are the indices of the dictionary elements with non-zero coefficients. That is, they are given by $S(\beta)$ where

$$S : \mathbb{R}^{P\times D} \to \binom{[P]}{D} \tag{26}$$

$$\beta \mapsto \{p \in [P] : \|\beta_{p\cdot}\| > 0\}. \tag{27}$$

IsometryPursuit(Matrix $X \in \mathbb{R}^{D\times P}$)

1: Normalize $X' = w(X)$
2: Optimize $\widehat{\beta} = \widehat{\beta}^{\ell_2}_{MBP}(X', I_D)$
3: **Output** $\widehat{S} = S(\widehat{\beta})$

3.5 Theory

The intuition behind our application of multitask basis pursuit is that submatrices consisting of vectors which are closer to 1 in length and more orthogonal will have smaller loss. A key theoretical assertion is that IsometryPursuit is invariant to choice of basis for X.

Proposition 17 *Let $U \in \mathbb{R}^{D\times D}$ be orthonormal. Then $S(\widehat{\beta}(UX)) = S(\widehat{\beta}(X))$.*

This proposition establishes the basis-independence of Isometry Pursuit and that we may replace I_D in the constraint by any orthonormal $D \times D$ matrix. It is also essential for showing the following main theoretical result.

When a rank D orthonormal column-submatrix $X_{\cdot S}$ exists, the output of Program 25 will contain S.

Proposition 18 *Let $X \in \mathbb{R}^{D\times P}$ have a rank D orthonormal column submatrix $X_{\cdot S}$. Then $S \subseteq \widehat{\beta}(X)$.*

Proofs of these propositions are given in Section 6.2, with corresponding experimental results in Section 4. The proof mostly relies on the KKT conditions [25] being used to verify that an isometric subset yields a corresponding basis pursuit solution for an appropriate normalized matrix. This is a formalization of our intuition that longer and more orthogonal vectors will minimize loss. We then apply a simple argument to show that the ℓ_2 minimizing solution is the least sparse solution, which means that it contains the isometry solution, should it exist.

3.6 Two-stage isometry pursuit

Proposition 18 suggests the following algorithm, which first uses the convex IsometryPursuit algorithm to prune the P candidate features and then applies brute search upon the reduced feature set. The proposition says that this approach is guaranteed to select an isometric submatrix should one exist. The two-stage approach is standard in the Lasso literature [26, 27, 10], and Proposition 18 justifies it in our application. This method forms our practical isometry estimator, the core advantage of which over purely brute search is a substantially improved computational complexity.

TwoStageIsometryPursuit(Matrix $X \in \mathbb{R}^{D\times P}$)

1: $\widehat{S}_{IP} = \text{IsometryPursuit}(X)$
2: $\widehat{S} = \text{BruteSearch}(X_{\cdot\widehat{S}_{IP}}, \ell)$
3: **Output** $\widehat{S}$

3.7 Non-uniqueness

The isometric solution is optimum among cardinality D solutions.

Proposition 19 $\ell(X_S) = D, |S| = D \Leftrightarrow X_S X_S^T = I_D$

However, there exist Isometry Pursuit solutions of minimum norm that are not isometries. For example, *Mercedes-Benz frame* matrix X achieves minimium norm.

Example 1 *Let* $D = 2, P = 3$. *Let* $X_S = \begin{bmatrix} 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}, \beta = \begin{bmatrix} 0 & \frac{2}{3} \\ \frac{1}{\sqrt{3}} & -\frac{1}{3} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{3} \end{bmatrix}$ *Then* $X\beta = I_D$ *and* $\|\beta\|_{1,2} = \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = 2 = D$. *Since the identity matrix is a solution regardless of* X_{-S}, *this is a non-isometric minimum loss solution.*

The upshot of this example is that IsometryPursuit can select solutions that are relatively non-isometric. The Mercedes-Benz frame is an example of a tight frame - a frame whose non-zero singular values are equal. While there are non-tight frames that have minimizing solutions, we do not here resolve whether there exist tight frames with unit norm columns that have minimum norm solutions.

3.8 Computation complexity

IsometryPursuit is more computationally efficient than BruteSearch search for the minimizer of ℓ. Interestingly, their asymptotic per-iteration complexities are the same.

Proposition 20 *The computational complexity* C_ℓ *of 10 is* $O(D^2P)$.

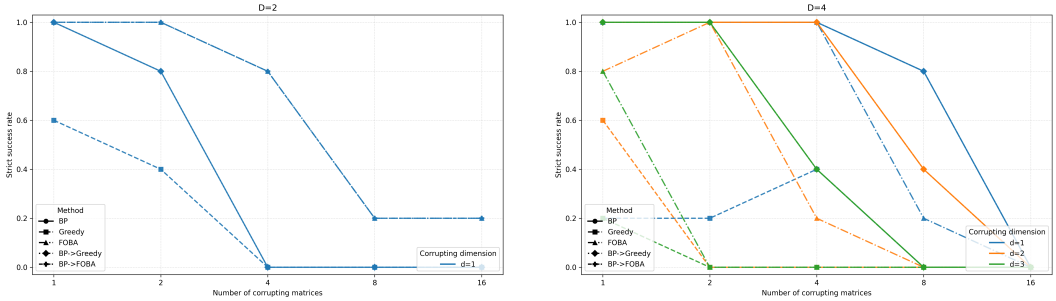
The above proposition follows immediately from the computational complexity of singular value decomposition [28]. The per-iteration complexity of IsometryPursuit is also limited by the computational complexity of singular value decomposition, albeit in the more intricate manner described in Section 6.2.7.

Proposition 21 *The computational complexity of each step of Isometry Pursuit is* $O(D^2P)$.

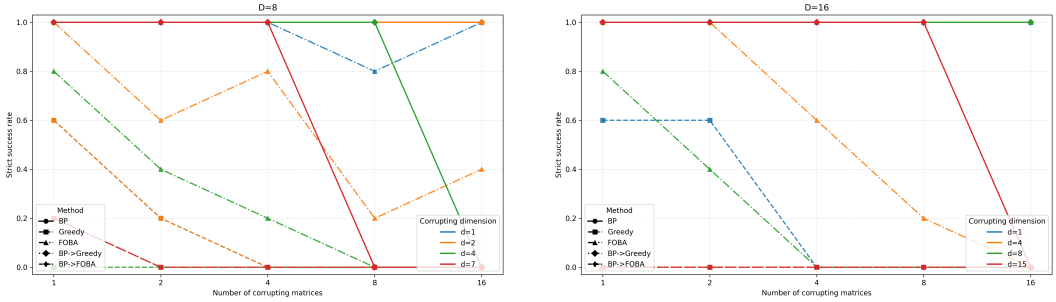
The difference therefore lies in the relation of the number of iterations required to achieve a given error tolerance with IsometryPursuit and BruteSearch. To optimize, we use the SCS interior point solver [29] from CVXPY [30, 31], which is able to push sparse values arbitrarily close to 0 [32]. SCS (aka ADMM) has asymptotic (in error) accuracy $\epsilon < O(\frac{1}{\mathcal{S}})$, where $\mathcal{S}$ is the number of iterations [33]. This means that the number of iterations required to achieve a given error tolerance scales inversely with the desired level of accuracy. We examine the effect of iteration number on the convergence to a sparse solution in Section 4.

4 Experiments

Say you are hosting a dinner party, and wish to select a balanced set of wines for drinking and flowers for decoration. We demonstrate TwoStageIsometryPursuit, full-dictionary forward-backward search, and IsometryPursuit-restricted forward-backward search on the Iris and Wine datasets [34, 35, 36]. This has an intuitive interpretation as selecting diverse elements that reflects the peculiar structure of the diversification problem. Features like *petal width* are rows in X. Thus, in diversification, $P = n$. Due to differences in



(a) $D = 2$ (b) $D = 4$



(c) $D = 8$ (d) $D = 16$

Figure 5: Synthetic identity-block experiments. Each matrix contains a planted identity block and lower-rank corrupting blocks. Success denotes exact recovery of the planted D columns.

Name	D	P	R	$\ell(X_{\cdot\widehat{S}_{FB}})$	$\lvert\widehat{S}_{IP}\rvert$	$\ell(X_{\cdot\widehat{S}_{FB\lvert IP}})$	$\widehat{P}(\ell(X_{\cdot\widehat{S}_{FB}}) > \ell(X_{\cdot\widehat{S}_{FB\lvert IP}}))$	$\widehat{P}(\ell(X_{\cdot\widehat{S}_{FB}}) = \ell(X_{\cdot\widehat{S}_{FB\lvert IP}}))$	Paired t-test p-value
Iris	4	75	25	7.3 ± 0.48	6.7 ± 1	7.2 ± 0.58	0.4	0.32	0.098
Wine	13	89	25	89 ± 6.7	26 ± 2.1	88 ± 6.4	0.28	0.68	0.0067
Ethanol	2	756	25	2.7 ± 0.97	2.8 ± 0.4	2.7 ± 0.98	0.12	0.6	0.47

Table 3: Experimental parameters and results. For Iris and Wine, probability estimates $\widehat{P}$ result from random downsampling features $[P]$ by a factor of 2 to create R replicates, while for Ethanol, replicates correspond to individual data points. Distributional probabilities $\widehat{P}(\ell(X_{\cdot\widehat{S}_{FB}}) > \ell(X_{\cdot\widehat{S}_{FB|IP}}))$ and $\widehat{P}(\ell(X_{\cdot\widehat{S}_{FB}}) = \ell(X_{\cdot\widehat{S}_{FB|IP}}))$ are empirical across replicates. Here $\widehat{S}_{FB}$ denotes the support selected by full-dictionary forward-backward search, and $\widehat{S}_{FB|IP}$ denotes the support selected by forward-backward search after restricting the dictionary to $\widehat{S}_{IP}$. The final column reports the one-sided paired t-test p-value for the alternative that full-dictionary forward-backward losses exceed restricted forward-backward losses.

scales between features, these are first standardized. For the Wine dataset, exhaustive BruteSearch search is prohibitive in $D = 13$, and so we compare practical discrete searches after random downsampling of candidate columns.

Our main interpretability dataset is the Ethanol dataset from Chmiela et al. [37], Koelle et al. [10]. For this dataset we evaluate a dictionary of interpretable features - bond torsions - for their ability to parameterize the molecular configuration space. In this interpretability use case, columns denote gradients of informative features. We compute Jacobian matrices of putative parametrization functions and project them onto estimated tangent spaces (see Koelle et al. [10] for preprocessing details). Rather than selecting between data points, we are selecting between functions which parameterize the data. Replicates are created by sampling from data points and their corresponding tangent spaces are estimated in $B = 252$. This dataset exemplifies our motivating example - the search for locally isometric embeddings.

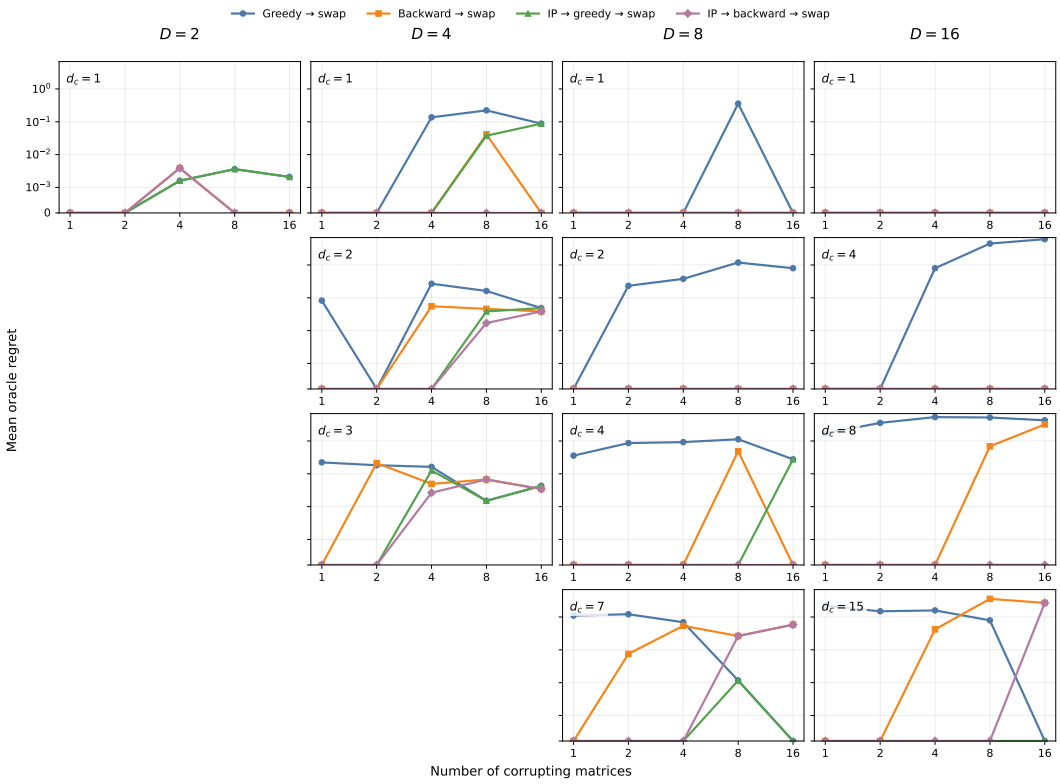


Figure 6: Mean oracle regret in the synthetic identity-block experiments. Columns give target dimension D and panels descend through the available corrupting dimensions d_c. Within each panel, the horizontal axis is the number of corrupting matrices and colors identify the four search methods. Each point averages five replicates at the displayed configuration.

Method	Wine ($P = 89$)	Iris ($P = 75$)	Ethanol ($P = 756$)
Greedy → swap	24.0% / 3.239	60.0% / 0.179	84.0% / 0.0188
Backward → swap	72.0% / 0.605	88.0% / 0.0170	96.0% / 0.00330
IsometryPursuit→ greedy → swap	40.0% / 2.446	56.0% / 0.0814	64.0% / 0.0180
	26.2/89 (29.4%)	6.72/75 (9.0%)	2.80/756 (0.37%)
IsometryPursuit→ backward → swap	64.0% / 0.615	64.0% / 0.0545	64.0% / 0.0180
	26.2/89 (29.4%)	6.72/75 (9.0%)	2.80/756 (0.37%)

Table 4: Oracle attainment and regret for the four discrete-search pipelines. Each entry is $\widehat{P}(\mathcal{R}_{m,r} = 0)$ / $\overline{\mathcal{R}}_m$: the percentage of replicates on which the method wins or ties for the lowest loss among the four methods, followed by its mean oracle regret. For the two IsometryPursuit pipelines, the second line reports $\overline{|\widehat{S}_{IP}|}/P$ and the corresponding retained percentage. Losses within 10^{-9} are treated as ties. Each dataset contains 25 replicates. Because loss scales differ across datasets, regret magnitudes should be compared within columns.

To compare all four practical searches directly, define the replicate-level oracle loss as $\ell_r^\star = \min_m \ell_{m,r}$ and the oracle regret of method m as $\mathcal{R}_{m,r} = \ell_{m,r} - \ell_r^\star$. Table 4 reports both the empirical probability of attaining the oracle loss and the mean oracle regret.

Figure 5 reports exact recovery rates in the synthetic identity-block experiments. Table 3 compares the final isometry losses from full-dictionary forward-backward search with losses from forward-backward search inside the IsometryPursuit support. Table 3 and Figure 7 detail intermediate support recovery cardinalities from IsometryPursuit. Isometry pursuit substantially reduced the size of the dictionary and made second-stage search computationally feasible. Wall-clock runtimes are given in Section 6.5.

5 Discussion

We have shown that multitask basis pursuit can help select isometric submatrices from appropriately normalized wide matrices. This approach - IsometryPursuit - is a convex alternative to greedy methods for selection of orthonormalized features from within a dictionary. Isometry pursuit can be applied to diversification and geometrically-faithful coordinate estimation. Our results exemplify these applications, but more can be done.

On the theoretical side, a critical unresolved question is the relation of the brute force and IsometryPursuit minimizers. Essentially, there are two things that can go wrong. First, IsometryPursuit may trim insufficiently for brute force search to be computationally feasible. Second, the recovered support may not contain the true minimizer. While we have ruled out the second problem in the case where an isometric solution does exist, in practice we often see greedy recovery defeat IsometryPursuit. Characterization of support recovery here is somewhat different from existing analysis of greedy [38, 39, 40, 41] and convex [18, 42, 43] basis pursuit formulations. Partly, this is because we use the ℓ_1 minimizer to distinguish between ℓ_0 optima rather than to identify them. Also, IsometryPursuit can generate non-unique solutions even when the design matrix satisfies general position, which is not true for the univariate formulation [44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54], but

The principal algorithmic variant of interest is the combination of the multitask lasso [25] extension of our estimator with D function selection within $\mathbb{R}^B$. First, introducing a Lasso penalty may ameliorate IsometryPursuit's reliance on the second stage of coordinate selection to complete the estimation process. Using the Lasso penalty, we can smoothly move between the multitask basis pursuit solution and more sparse solutions [23]. Second, we often may want to select some orthonormal subset of dimension much less than D. Finally, response matrices may be extended beyond the identity.

These extensions may faciliate an interesting set of applications diversification in recommendation [55, 56, 57] and retrieval [58, 59, 60, 61, 62]. Maximum cosine similarity used in retrieval corresponds to the first coefficient of the Lasso regularization path [10]. Greedy algorithms predominate in diversification [63, 64, 65, 66, 67, 68, 69, 70, 71, 59, 2, 72], and convex reformulations have been studied to add speed and convergence to a global minima.

Compared with Sparse PCA [73, 74, 75, 76], we seek a low-dimensional representation from the set of these column vectors rather than their span.

Diversification.

While other applications like diversification are possible and potentially , the embedding

Diversification may be framed an a kind of orthogonal selection problem. Cosine similarity Colbert

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6 Supplement

This section contains algorithms, proofs, and experiments in support of the main text.

6.1 Algorithms

We give definitions of the brute and greedy algorithms for the combinatorial problem studied in this paper. The brute force algorithm is computationally intractable for all but the smallest problems, but always finds the global minima.

BruteSearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f)

for each combination $S \subseteq \{1, 2, \ldots, P\}$ with $|S| = D$ **do**
  Evaluate $f(X_{.S})$
end for
Output the combination S^* that minimizes $f(X_{.S})$

Greedy algorithms are computationally expedient but can get stuck in local optima [77, 78], even with randomized restarts [79].

GreedySearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f)

Initialize $S = \emptyset$
while $|S| < D$ **do**
  Initialize $S_{\text{best}} = S$
  Initialize $f_{\text{best}} = \infty$
  for each $p \in \{1, 2, \ldots, P\} \setminus S$ **do**
    Evaluate $f(X_{.(S \cup \{p\})})$
    if $f(X_{.(S \cup \{p\})}) < f_{\text{best}}$ **then**
      Update $S_{\text{best}} = S \cup \{p\}$
      Update $f_{\text{best}} = f(\mathcal{X}_{.(S \cup \{p\})})$
    end if
  end for
  Update $S = S_{\text{best}}$
end while
Return S

BackwardGreedySearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f)

Initialize $S = \{1, 2, \ldots, P\}$
while $|S| > D$ **do**
  Initialize $S_{\text{best}} = S$
  Initialize $f_{\text{best}} = \infty$
  for each $p \in S$ **do**
    Evaluate $f(X_{.(S \setminus \{p\})})$
    if $f(X_{.(S \setminus \{p\})}) < f_{\text{best}}$ **then**
      Update $S_{\text{best}} = S \setminus \{p\}$
      Update $f_{\text{best}} = f(\mathcal{X}_{.(S \setminus \{p\})})$
    end if
  end for
  Update $S = S_{\text{best}}$
end while
Return S

ForwardBackwardSearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f)

Initialize $S \leftarrow$ GreedySearch(X, f)
Initialize $f_{\text{current}} \leftarrow f(X_{.S})$
for $p = 1, 2, \ldots, P$ **do**
  Initialize $S_{\text{best}} \leftarrow S$
  Initialize $f_{\text{best}} \leftarrow f_{\text{current}}$
  for each $j \in \{1, 2, \ldots, P\} \setminus S$ **do**
    for each $q \in S$ **do**
      Initialize $S_{\text{candidate}} = (S \cup \{j\}) \setminus \{q\}$
      Evaluate $f(X_{.S_{\text{candidate}}})$
      if $f(X_{.S_{\text{candidate}}}) < f_{\text{best}}$ **then**
        Update $S_{\text{best}} \leftarrow S_{\text{candidate}}$
        Update $f_{\text{best}} \leftarrow f(X_{.S_{\text{candidate}}})$
      end if
    end for
  end for
  if $S_{\text{best}} = S$ **then**
    Break
  end if
  Update $S \leftarrow S_{\text{best}}$
  Update $f_{\text{current}} \leftarrow f_{\text{best}}$
end for
Return S

6.2 Proofs

6.2.1 Proof of Proposition 6

Proof: First,

$$\frac{1}{2}\log\det(X^TX) = \frac{1}{2}\log\prod_{d=1}^{D}\sigma_d(X)^2 \tag{28}$$

$$= \log\prod_{d=1}^{D}\sigma_d(X) \tag{29}$$

$$= \sum_{d=1}^{D}\log\sigma_d(X). \tag{30}$$

Secondly, I_D and X^TX are diagonalizable in the same basis, so

$$\log\det(\frac{X^TX + I_D}{2}) = \log\prod_{d=1}^{D}\frac{\sigma_d(X)^2+1}{2} \tag{31}$$

$$= \sum_{d=1}^{D}\log\frac{\sigma_d(X)^2+1}{2}. \tag{32}$$

Then,

$$D(X^TX, I_D)^2 = \sum_{d=1}^{D}(\log\frac{\sigma_d(X)^2+1}{2} - \log\sigma_d(X)) \tag{33}$$

$$= \sum_{d=1}^{D}\log\frac{\sigma_d(X)^2+1}{2\sigma_d(X)} \tag{34}$$

$$= \sum_{d=1}^{D}\log\frac{\sigma_d(X)+\sigma_d(X)^{-1}}{2}. \tag{35}$$

□

6.2.2 Singular values of nearly colinear vectors

We now exhibit a matrix for which no g can be chosen such that a log det divergence satisfies Properties **??** and **??**. We parametrize this matrix as

$$X_\rho(r) = \begin{bmatrix} r & \rho & \rho \\ 0 & \sqrt{1-\rho^2} & \frac{\rho(1-\rho)}{\sqrt{1-\rho^2}} \\ 0 & 0 & \sqrt{\frac{(1-\rho)(1+2\rho)}{1+\rho}} \end{bmatrix} \tag{36}$$

where r is the length of the second vector and ρ controls the colinearity between vectors. We compute

$$\frac{\partial \ell(r)s}{\partial r}|_{r=1} = \sum_{d=1}^{3} g'(\sigma_d(X_\rho(r))) \frac{\partial}{\partial r} \sigma_d(X_\rho(r)) \Big|_{r=1}. \tag{37}$$

We will to show that, for any log-symmetric g such that $g(t) = g(1/t)$ and any $\rho \in (-1, 1)$

$$\frac{\partial \ell(r)}{\partial r}|_{r=1} \neq 0. \tag{38}$$

First, let's compute $\sigma_d(X_\rho(r))$.

$$X_\rho(r)^T X_\rho(r) = \begin{bmatrix} r^2 & r\rho & r\rho \\ r\rho & 1 & \rho \\ r\rho & \rho & 1 \end{bmatrix}. \tag{39}$$

Recall that λ is an eigenvalue of a matrix A if there is a non-zero vector v such that $Av = \lambda v$. Equivalently,

$$\det(A - \lambda I) = 0. \tag{40}$$

In our case, this gives

$$\begin{aligned}
0 &= \det \begin{bmatrix} r^2-\lambda & r\rho & r\rho \\ r\rho & 1-\lambda & \rho \\ r\rho & \rho & 1-\lambda \end{bmatrix} && (41)\\
&= (r^2-\lambda)\left((1-\lambda)^2 - \rho^2\right) - r\rho\left(r\rho(1-\lambda) - r\rho^2\right) + r\rho\left(r\rho^2 - r\rho(1-\lambda)\right) && (42)\\
&= (r^2-\lambda)\left((1-\lambda)^2 - \rho^2\right) - 2r^2\rho^2(1-\lambda) + 2r^2\rho^3 && (43)\\
&= (r^2-\lambda)(1-\rho-\lambda)(1+\rho-\lambda) - 2r^2\rho^2(1-\rho-\lambda) && (44)\\
&= (1-\rho-\lambda)\left((r^2-\lambda)(1+\rho-\lambda) - 2r^2\rho^2\right) && (45)\\
&= (1-\rho-\lambda)\left(r^2(1+\rho) - r^2\lambda - \lambda(1+\rho) + \lambda^2 - 2r^2\rho^2\right) && (46)\\
&= (1-\rho-\lambda)\left(\lambda^2 - (r^2+1+\rho)\lambda + r^2(1+\rho-2\rho^2)\right). && (47)
\end{aligned}$$

Thus the eigenvalues of $X_\rho(r)^T X_\rho(r)$ are

$$\begin{aligned}
\lambda_1(r) &= 1-\rho, && (48)\\
\lambda_2(r) &= \frac{r^2+1+\rho+\sqrt{(r^2+1+\rho)^2 - 4r^2(1+\rho-2\rho^2)}}{2} && (49)\\
&= \frac{r^2+1+\rho+\sqrt{\left(r^2-(1+\rho)\right)^2 + 8r^2\rho^2}}{2}, && (50)\\
\lambda_3(r) &= \frac{r^2+1+\rho-\sqrt{(r^2+1+\rho)^2 - 4r^2(1+\rho-2\rho^2)}}{2} && (51)\\
&= \frac{r^2+1+\rho-\sqrt{\left(r^2-(1+\rho)\right)^2 + 8r^2\rho^2}}{2}. && (52)
\end{aligned}$$

Since singular values are the square roots of the eigenvalues of $X_\epsilon(r)^T X_\epsilon(r)$,

$$\sigma_1(X_\rho(r)) = [1-\rho]^{1/2}, \tag{53}$$

$$\sigma_2(X_\rho(r)) = \left[\frac{r^2+1+\rho+\sqrt{\left(r^2-(1+\rho)\right)^2+8r^2\rho^2}}{2}\right]^{1/2}, \tag{54}$$

$$\sigma_3(X_\rho(r)) = \left[\frac{r^2+1+\rho-\sqrt{\left(r^2-(1+\rho)\right)^2+8r^2\rho^2}}{2}\right]^{1/2}. \tag{55}$$

Differentiating the singular values gives

$$\frac{\partial\sigma_1}{\partial r} = 0, \tag{56}$$

$$\frac{\partial\sigma_2}{\partial r} = \frac{1}{2}\left(\frac{r^2+1+\rho+\sqrt{(r^2-(1+\rho))^2+8r^2\rho^2}}{2}\right)^{-1/2} \tag{57}$$

$$\times \frac{2r+\frac{1}{2}\left(\left((r^2-(1+\rho))^2+8r^2\rho^2\right)^{-1/2}\left(2(r^2-(1+\rho))(2r)+16r\rho^2\right)\right)}{2}, \tag{58}$$

$$\frac{\partial\sigma_3}{\partial r} = \frac{1}{2}\left(\frac{r^2+1+\rho-\sqrt{(r^2-(1+\rho))^2+8r^2\rho^2}}{2}\right)^{-1/2} \tag{59}$$

$$\times \frac{2r-\frac{1}{2}\left(\left((r^2-(1+\rho))^2+8r^2\rho^2\right)^{-1/2}\left(2(r^2-(1+\rho))(2r)+16r\rho^2\right)\right)}{2}. \tag{60}$$

Then

$$\sigma_1(X_\rho(1)) = (1-\rho)^{1/2}, \tag{61}$$

$$\sigma_2(X_\rho(1)) = (1+2\rho)^{1/2}, \tag{62}$$

$$\sigma_3(X_\rho(1)) = (1-\rho)^{1/2}, \tag{63}$$

$$\left.\frac{\partial\sigma_1}{\partial r}\right|_{r=1} = 0, \tag{64}$$

$$\left.\frac{\partial\sigma_2}{\partial r}\right|_{r=1} = \frac{1}{3}(1+2\rho)^{1/2}, \tag{65}$$

$$\left.\frac{\partial\sigma_3}{\partial r}\right|_{r=1} = \frac{2}{3}(1-\rho)^{1/2}. \tag{66}$$

Note that in fact σ_2 is the largest singular value, but for notational consistency we won't reorder.

Now we can specify g. Compute

$$\frac{\partial\ell(r)}{\partial r}|_{r=1} = g'\left((1-\rho)^{1/2}\right)\cdot 0 + g'\left((1+2\rho)^{1/2}\right)\frac{1}{3}(1+2\rho)^{1/2} + g'\left((1-\rho)^{1/2}\right)\frac{2}{3}(1-\rho)^{1/2} \tag{67}$$

$$= \frac{1}{3}g'\left((1+2\rho)^{1/2}\right)(1+2\rho)^{1/2} + \frac{2}{3}g'\left((1-\rho)^{1/2}\right)(1-\rho)^{1/2}. \tag{68}$$

Suppose $\frac{\partial\ell(r)}{\partial r}|_{r=1} = 0$. Then

$$\frac{1}{3}g'\left((1+2\rho)^{1/2}\right)(1+2\rho)^{1/2} = -\frac{2}{3}g'\left((1-\rho)^{1/2}\right)(1-\rho)^{1/2}. \tag{69}$$

What is a suitable g? We can rewrite the above constraint as

$$g'\left((1+2\rho)^{1/2}\right) = -2\left(\frac{1-\rho}{1+2\rho}\right)^{1/2} g'\left((1-\rho)^{1/2}\right). \tag{70}$$

Let $\rho = 1/2$. Then

$$g'(\sqrt{2}) = -2\left(\frac{1}{4}\right)^{1/2} g'\left((1/2)^{1/2}\right) \tag{71}$$

$$= -g'(1/\sqrt{2}) \tag{72}$$

However, by assumption, we have $g(t) = g(1/t)$, and so $g'(t) = \frac{1}{t^2}g'(1/t)$. At $t = \sqrt{2}$, $g'(\sqrt{2}) = \frac{1}{2}g'(1/\sqrt{2})$. Thus $\frac{1}{2}g'(1/\sqrt{2}) = -g'(1/\sqrt{2})$ and so $g'(1/\sqrt{2})$ and g is not monotone.

6.2.3 Proof of Proposition 12

Proof: The LogSumExp relaxation of $\max(t, t^{-1})$ is

$$\mathrm{LSE}(t, t^{-1}) = \log\left(e^t + e^{t^{-1}}\right). \tag{73}$$

The exponential of this relaxation is

$$\exp\left(\mathrm{LSE}(t, t^{-1})\right) = \exp\left(\log\left(e^t + e^{t^{-1}}\right)\right) \tag{74}$$

$$= e^t + e^{t^{-1}}. \tag{75}$$

Since $g(1) = 1$, its inverse is normalized to equal 1 at $t = 1$:

$$\frac{1}{g(t)} = \frac{e^t + e^{t^{-1}}}{2e} \tag{76}$$

$$= \exp\left(\mathrm{LSE}(t, t^{-1}) - 1 - \log 2\right), \tag{77}$$

which is the normalized exponential of the LogSumExp relaxation. □

6.2.4 Proof of Proposition 17

In this proof we first show that the penalty $\|\beta\|_{1,2}$ is unchanged by unitary transformation of β.

Proposition 22 *Let $U \in \mathbb{R}^{D\times D}$ be unitary. Then $\|\beta\|_{1,2} = \|\beta U\|$.*

Proof:

$$\|\beta U\|_{1,2} = \sum_{p=1}^{P} \|\beta_{p.} U\| \tag{78}$$

$$= \sum_{p=1}^{P} \|\beta_{p.}\| \tag{79}$$

$$= \|\beta\|_{1,2} \tag{80}$$

□

We then show that this implies that the resultant loss is unchanged by unitary transformation of X.

Proposition 23 *Let $U \in \mathbb{R}^{D\times D}$ be unitary. Then $\widehat{\beta}(UX) = \widehat{\beta}(X)U$.*

Proof:

$$\widehat{\beta}(UX) = \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = UX\beta \tag{81}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ U^{-1}U = U^{-1}UX\beta U \tag{82}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = X\beta U \tag{83}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta U\|_{1,2} \ : \ I_D = X\beta U \tag{84}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = X\beta. \tag{85}$$

□

6.2.5 Proof of Proposition 18

In this proof we first show that the orthonormal submatrix $X_{.S}$ is contained within the set $\widehat{\beta}_{MBP}(w(X), I_D)$.

Proposition 24 *Let $X_{.S}$ be a rank D orthonormal submatrix of X. Then $S \in S(\beta) : \beta \in \widehat{\beta}_{MBP}(w(X), I_D))$.*

Proof: By Proposition 17 without loss of generality, let $X_{.S} = I_D$. Also, without loss of generality let $S = \{1 \dots D\}$. We show that $\beta = [I_D \; 0]$ satisfies the KKT conditions for $X = [I_D \; X_{-S}]$.

Recall that the KKT conditions for a convex optimization problem are a set of conditions on an element of the domain of the optimization algorithm that, if satisfied, certify that the element is a solution. To write out the KKT conditions, we define the Lagrangian

$$\mathcal{L}(X, \beta, \nu) = \|\beta\|_{1,2} + \nu^T (I_D - w(X)\beta) \tag{86}$$

where $\nu \in \mathbb{R}^{D \times D}$ is the dual variable.

The KKT conditions are then

- Primal feasibility: $w(X)\beta = I_D$
- Stationarity: there exists a dual variable ν such that $0 \in \partial \|\beta\|_{1,2} - w(X)^T \nu$ where ∂ is the subdifferential operator.

where dual feasibility and complementary slackness are ignorable by virtue of the absence of inequality constraints.

First note that in our case, primal feasibility is satisfied by X being rank D and w not effecting the rank of the transformed matrix since it only rescales length and not direction.

For stationarity, recall that

$$\|\beta\|_{1,2} = \sum_{p=1}^{P} \|\beta_{p.}\|_2. \tag{87}$$

Then, by the definition of subdifferential

$$\partial \|\beta\|_{1,2} = \begin{bmatrix} I_D \\ v_{d+1} \\ \dots \\ v_P \end{bmatrix} \tag{88}$$

where $v_p \in \mathbb{R}^D$ can be any vector satisfying $\|v_p\|_2 \leq 1$.

We therefore must show that there exists a ν that satisfies

$$\begin{bmatrix} I_D \\ w(X_{-S})^T \end{bmatrix} \nu = \begin{bmatrix} I_D \\ V_{-S} \end{bmatrix}. \tag{89}$$

At this point we can see that in fact I_D is an appropriate choice of ν since normalization by w leads to vectors satisfying $\|w(X)\|_2 \leq 1$.

Since for this β, $S(\beta) = S$, we have shown that S is the support of a solution to $\beta_{MBP}(w(X), I_D)$. □

The next part of the proof handles the presence of non-unique solutions.

Proposition 25 $S(\widehat{\beta}(X)) = \cup \, S(\beta) \; : \; \beta \in \beta_{MBP}(w(X), I_D)$

Proof: The proposition follows from geometric properties of the solution set. We ignore the case for singleton solutions, for which the proposition is self-evident. Recall that $\widehat{\beta}(X) := \widehat{\beta}^{\ell_2}_{MBP}(w(X), I_D) = \arg\min_{\beta \in \mathbb{R}^{P \times D}} \|\beta\|_F : \beta \in \widehat{\beta}_{MBP}(X, I_D)$. By definition of

$\widehat{\beta}_{MBP}(X, I_D)$, $\|\beta_{p\cdot}\|_2 : \beta \in \widehat{\beta}_{MBP}(X, I_D)$ is an affine linear subspace of $\mathbb{R}^P$. The projection of the vector of minimum ℓ_2 distance from the origin to a affine space intersects that space perpendicularly. However, for all $\|\beta_{\cdot p}\|_2$ such that β is on the intersection of this affine space with the coordinate hyperplanes formed by fixing any choices of $\|\beta_{\cdot p}\|_2 = 0$, this vector doesn't intersect this space perpendicularly. Thus, $\widehat{\beta}(X)$ is not on the intersection of the solution polytope $\widehat{\beta}_{MBP}(w(X), I_D)$ with the coordinate hyperplanes. It is therefore maximally non-sparse with respect to any of the minimizing solutions. □

6.2.6 Proof of Proposition 19

X_S admits a QR decomposition $X_S = QR$ where Q is orthogonal and R is upper triangular. By Proposition 17, we can assume without loss of generality that $Q = I_D$. That is, X_S is upper triangular. Thus, $\beta = X_S^{-1}$ is upper triangular with diagonal elements $\beta_{dd} = R_{dd}^{-1}$. $\|\beta_{d\cdot}\|_2 \geq \beta_{dd}$, with equality only occurring when the off-diagonal elements are 0. Therefore, $\|\beta_{1,2}\| \geq D$ with equality occurring only when $\beta = I_D$.

6.2.7 Proof of Proposition 21

Proof: The augmented split Lagrangian minimized by ADMM is

$$\mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z) = \|z\|_{1,2} + \text{tr}(\nu^T(I_D - w(X)\beta)) + \rho\|I_D - w(X)\beta\|_F^2 + \gamma\|\beta - z\|_F^2 \tag{90}$$

This minimization problem is

$$\arg \min_{\beta, z \in P \times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z) \tag{91}$$

For a more detailed explanation of the reasons to prefer optimizing the split augmented Lagrangian over direct optimization of Equation 21 see Boyd [80].

The three main subroutines (95 - 99) called by ADMM determine the per iteration complexity. The update rules of the ADMM algorithm used to optimize the augmented split Lagrangian are

$$\beta^{k+1} = \arg \min_{\beta \in P \times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z^k) \tag{92}$$

$$z^{k+1} = \arg \min_{z \in P \times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta^{k+1}, z) \tag{93}$$

$$\nu^{k+1} = \nu^k + \rho(I_D - w(X)\beta^{k+1}) \tag{94}$$

We will analyze the computational complexity of each of these steps by rewriting the update rules according to the specifics of the isometry pursuit problem.

The beta update Equation 95 projects the solution onto the feasible set.

$$\beta^{k+1} = \arg \min_{\beta \in P \times D} \text{tr}({\nu^k}^T(I_D - w(X)\beta)) + \rho\|I_D - w(X)\beta\|_2^2 + \gamma\|\beta - z^k\|_2^2 \tag{95}$$

$$= (\rho\, w(X)^T w(X) + \gamma I_D)^{-1}(\rho\, w(X)^T I_D + \gamma z^k + w(X)^T \nu^k) \tag{96}$$

The computational complexity of this computation is dominated by the inverse - that is, $O(D^2P)$.

Subsequently, the z update Equation 97 is soft-thresholding operation onto a contracted level set $\|\|_{1,2}$ ball.

$$z^{k+1} = \arg \min_{z \in P \times D} \|z\|_{1,2} + \gamma\|z - \beta^{k+1}\|_2^2 \tag{97}$$

$$= \begin{cases} (1 - \frac{1}{\gamma\|\beta_{j\cdot}^{k+1}\|_2})(\beta_{j\cdot}^{k+1}) \text{ if } \|\beta_{j\cdot}^{k+1}\|_2 > \frac{1}{\gamma} \\ 0 \text{ otherwise} \end{cases} \tag{98}$$

The computational complexity of this operation is $O(PD)$.

Finally, the ν update Equation 99 increases the squeezing of the z and β variables towards to each other.

$$\nu^{k+1} = \nu^k + \rho(I_D - w(X)\beta^{k+1}) \tag{99}$$

In this step, the dual variable ν that controls the penalty accrued by the splitting is updated in the direction of the constraint violation. The computational complexity of this operation is dominated by that of the matrix multiplication $w(X)\beta^{k+1}$. That is, it is $O(D^2P)$. □

6.3 Partial derivatives of candidate losses

We summarize the column-wise derivatives of the losses in Table 2. Throughout this section let $X = [x_1, \dots, x_D] \in \mathbb{R}^{D\times D}$ be full rank. For a scalar loss $L(X)$, write

$$\nabla_X L = \left[\frac{\partial L}{\partial x_1}, \dots, \frac{\partial L}{\partial x_D}\right]. \tag{100}$$

All formulas are stated on smooth strata: rank deficient matrices, repeated singular values, and zero rows of pseudoinverse factors should be handled using subgradients.

We will use two common identities. First, if $X = U\,\mathrm{diag}(\sigma)V^\top$ is a singular value decomposition and $L(X) = \sum_{d=1}^{D} \phi(\sigma_d(X))$, then

$$\nabla_X L = U\,\mathrm{diag}(\phi'(\sigma_1), \dots, \phi'(\sigma_D))V^\top. \tag{101}$$

Second, for normalized losses let $N = n_q^{\times D}(X)$. If $M = \nabla_N L$, then the column-wise chain rule is

$$\frac{\partial L}{\partial x_p} = \mathrm{D}n_q(x_p)^\top M_{\cdot p}. \tag{102}$$

For $r = \|x\|_2$, write $n_q(x) = s(r)x$ with $s(r) = q(r)/r$. Then

$$\mathrm{D}n_q(x) = s(r)I_D + \frac{s'(r)}{r}xx^\top, \tag{103}$$

$$s'(r) = \frac{q'(r)}{r} - \frac{q(r)}{r^2}, \tag{104}$$

$$q'(r) = -\frac{2e(e^r - r^{-2}e^{1/r})}{(e^r + e^{1/r})^2}. \tag{105}$$

Since $\mathrm{D}n_q(x)$ is symmetric, the transpose in this chain rule may be omitted.

Distortion. Let $\sigma_1 = \sigma_{\max}(X)$ and $\sigma_D = \sigma_{\min}(X)$ with corresponding singular vectors u_1, v_1 and u_D, v_D. For

$$L_{\mathrm{dist}}(X) = \frac{\sigma_1}{\sigma_D} - 1, \tag{106}$$

The spectral gradient identity gives

$$\nabla_X L_{\mathrm{dist}} = \frac{u_1 v_1^\top}{\sigma_D} - \frac{\sigma_1}{\sigma_D^2}u_D v_D^\top. \tag{107}$$

The derivative is insensitive to uniform rescaling, which is the sense in which this loss does not prefer unit length.

Isometry loss. For

$$L_{\mathrm{iso}}(X) = \sum_{d=1}^{D} g(\sigma_d(X)) - D, \tag{108}$$

we have

$$g'(t) = \frac{e^t - t^{-2}e^{1/t}}{2e}. \tag{109}$$

Thus

$$\nabla_X L_{\mathrm{iso}} = U\,\mathrm{diag}(g'(\sigma_1), \dots, g'(\sigma_D))V^\top. \tag{110}$$

S-divergence to identity. Let $G = X^\top X$ and

$$L_{\mathrm{sdiv}}(X) = \log\det\left(\frac{G + I_D}{2}\right) - \frac{1}{2}\log\det(G). \tag{111}$$

The constant $\log\det(1/2)$ does not contribute to the derivative. Using $d\log\det A = \mathrm{tr}(A^{-1}dA)$ and $dG = dX^\top X + X^\top dX$,

$$\nabla_X L_{\mathrm{sdiv}} = 2X(G + I_D)^{-1} - XG^{-1}. \tag{112}$$

Normalized log-det volume. Let $N = n_q^{\times D}(X)$ and

$$L_{\mathrm{nld}}(X) = -\sum_{d=1}^{D} \log \sigma_d(N). \tag{113}$$

For square full-rank N, this is $-\log|\det N|$, so

$$\nabla_N L_{\mathrm{nld}} = -N^{-\top}. \tag{114}$$

Therefore, by the column-wise chain rule,

$$\frac{\partial L_{\mathrm{nld}}}{\partial x_p} = -\mathrm{D}n_q(x_p)^\top (N^{-\top})_{\cdot p}. \tag{115}$$

Frobenius pseudoinverse. Let $N = n_q^{\times D}(X)$ and $B = N^{-1}$. For

$$L_{\mathrm{frob}}(X) = \|N^{-1}\|_F - \sqrt{D}, \tag{116}$$

the differential identity $dB = -B(dN)B$ gives

$$\nabla_N L_{\mathrm{frob}} = -\frac{N^{-\top} N^{-1} N^{-\top}}{\|N^{-1}\|_F}. \tag{117}$$

Thus

$$\frac{\partial L_{\mathrm{frob}}}{\partial x_p} = \mathrm{D}n_q(x_p)^\top \left(-\frac{N^{-\top} N^{-1} N^{-\top}}{\|N^{-1}\|_F} \right)_{\cdot p}. \tag{118}$$

Isometry pursuit proxy. Let $N = n_q^{\times D}(X)$ and $B = N^{-1}$. The differentiable proxy for the IsometryPursuit table row is

$$L_{\mathrm{ip}}(X) = \|N^{-1}\|_{1,2} - D. \tag{119}$$

Define $R \in \mathbb{R}^{D \times D}$ by

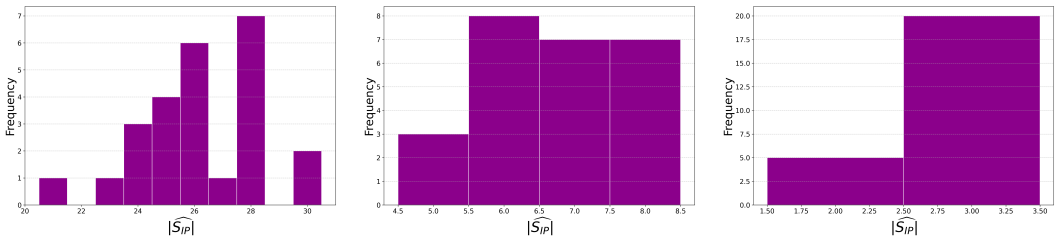
$$R_{p\cdot} = \frac{B_{p\cdot}}{\|B_{p\cdot}\|_2}. \tag{120}$$

Then $R \in \partial \|B\|_{1,2}$ on the smooth stratum where every row of B is nonzero, and

$$\nabla_N L_{\mathrm{ip}} = -B^\top R B^\top. \tag{121}$$

The column-wise gradient is therefore

$$\frac{\partial L_{\mathrm{ip}}}{\partial x_p} = \mathrm{D}n_q(x_p)^\top (-B^\top R B^\top)_{\cdot p}. \tag{122}$$



(a) Wine (b) Iris (c) Ethanol

Figure 7: Support cardinalities for each dataset after IsometryPursuit.

6.4 Support cardinalities

Figure 7 plots the distribution of $|\widehat{S}_{IP}|$ from Table 3 in order to contextualize the reported means. While typically $|\widehat{S}_{IP}| << P$, there are cases for Ethanol where this is not the case that drive up the means. Further investigation into the extreme failure of isometry pursuit to sparsify in the the minor chunk of the bimodal Ethanol support cardinality distribution is warranted.

6.5 Timing

While wall-time of algorithms is a non-theoretical quantity that depends on implementation details, it provides valuable context for practitioners. We therefore report the following runtimes on a 2021 Macbook Pro. The particularly high variance for brute force search in the second step of TwoStageIsometryPursuit is likely due to the large cardinalities reported in Figure 7.

Name	IP	2nd stage	Greedy
Iris	1.24 ± 0.02	0.00 ± 0.00	0.02 ± 0.00
Wine	2.32 ± 0.17	0.13 ± 0.12	0.03 ± 0.00
Ethanol	8.38 ± 0.57	0.55 ± 1.08	0.07 ± 0.01

Table 5: Algorithm runtimes in seconds across replicates.

Example 2

$$X_S = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} & \frac{-1}{2} \\ \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & 0 & 0 & & \\ 0 & 0 & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 00 & 0 & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & \end{bmatrix}. \tag{123}$$

Proof: Let

By Given the symmetry of the second and third columns around the y axis, the loss will not change regardless of whether we upweight one or downweight the other, these changes necessarily being in conjunction. We therefore assume that their loadings with respect to the second column of the target matrix (the D dimensional identity) are the same. Clearly, they will both then have value of $\frac{1}{\sqrt{3}}$. The question then becomes, what should the values of the first column in the loading matrix β be?

We can answer this question with a simple differential equation. Let b denote the constraint violation $1 - X_{1.}\beta_{.1}$.

Then

$$\frac{\partial b}{\partial \beta_{11}} = \frac{\partial b}{\partial \beta_{21}} \tag{124}$$

$$= \frac{1}{2} \tag{125}$$

On the other hand, due to the value of $\frac{1}{\sqrt{3}}$ already assumed for the second loading of the second and third features,

$$\frac{\partial l}{\partial \beta_{11}} = \frac{\partial l}{\partial \beta_{21}} \tag{126}$$

$$= \frac{\partial}{\partial x}\sqrt{\frac{1}{3} + x^2} \tag{127}$$

$$= \frac{1}{2}(\frac{1}{3} + x^2)^{-\frac{1}{2}}(2x) \tag{128}$$

$$= \frac{x}{\sqrt{\frac{1}{3} + x^2}}. \tag{129}$$

This means that the tradeoff

$$\frac{\partial l}{\partial b} = \frac{1}{2x}\sqrt{\frac{1}{3} + x^2}. \tag{130}$$

In contrast, $\frac{\partial l}{\partial \beta_{11}} = 1$ and similarly for its constraint violation. Solving for the equilibrium, we get

$$\frac{1}{2x}\sqrt{\frac{1}{3} + x^2} = 1 \tag{131}$$

$$\frac{1}{3} + x^2 = 4x^2 \tag{132}$$

$$\frac{1}{3} = 3x^2 \tag{133}$$

$$x = \frac{1}{\sqrt{3}}. \tag{134}$$

This leads to a value for l of

$$\sqrt{\frac{1}{3} + \frac{1}{9}} + \sqrt{\frac{1}{3} + \frac{1}{9}} + \frac{2}{3} \tag{135}$$

$$= \sqrt{\frac{4}{9}} + \sqrt{\frac{4}{9}} + \frac{2}{3} \tag{136}$$

$$= \frac{2}{3} + \frac{2}{3} + \frac{2}{3} \tag{137}$$

□